Double Spherical Pendulum Derivations

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1 Find the Lagrangian and the four Lagrange Equations.

1.1 Write down the kinetic and potential energies and hence the Lagrangian $\mathcal{L} = T - U$, using any convenient inertial reference frame.

Kinetic energy:

$$T = \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2 + \dot{z}_1^2) + \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2 + \dot{z}_2^2)$$

Potential energy[1]:

$$U = -m_1 g z_1 - m_2 g z_2$$

The Lagrangian:

$$\mathcal{L} = \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2 + \dot{z}_1^2) + \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2 + \dot{z}_2^2) + m_1 g z_1 + m_2 g z_2$$

1.2 Choose a convenient set of n generalized coordinates $q_1, \cdots, q_n$ and find expression for the original coordinates of step 1 in terms of your chosen generalized coordinates.

Generalized coordinates: $\{\theta_1, \theta_2, \phi_1, \phi_2\}$
Cartesian coordinates in terms of the generalized coordinates:

$$\begin{aligned}
x_1 &= R_1 \sin\theta_1 \cos\phi_1 \\
y_1 &= R_1 \sin\theta_1 \sin\phi_1 \\
z_1 &= R_1 \cos\theta_1 \\
x_2 &= R_1 \sin\theta_1 \cos\phi_1 + R_2 \sin\theta_2 \cos\phi_2 \\
y_2 &= R_1 \sin\theta_1 \sin\phi_1 + R_2 \sin\theta_2 \sin\phi_2 \\
z_2 &= R_1 \cos\theta_1 + R_2 \cos\theta_2
\end{aligned}$$

Velocities:

$$\begin{aligned}
\dot{x}_1 &= R_1(\dot{\theta}_1 \cos\theta_1 \cos\phi_1 - \dot{\phi}_1 \sin\theta_1 \sin\phi_1) \\
\dot{y}_1 &= R_1(\dot{\theta}_1 \cos\theta_1 \sin\phi_1 + \dot{\phi}_1 \sin\theta_1 \cos\phi_1) \\
\dot{z}_1 &= -R_1\dot{\theta}_1 \sin\theta_1 \\
\dot{x}_2 &= R_1(\dot{\theta}_1 \cos\theta_1 \cos\phi_1 - \dot{\phi}_1 \sin\theta_1 \sin\phi_1) + R_2(\dot{\theta}_2 \cos\theta_2 \cos\phi_2 - \dot{\phi}_2 \sin\theta_2 \sin\phi_2) \\
\dot{y}_2 &= R_1(\dot{\theta}_1 \cos\theta_1 \sin\phi_1 + \dot{\phi}_1 \sin\theta_1 \cos\phi_1) + R_2(\dot{\theta}_2 \cos\theta_2 \sin\phi_2 + \dot{\phi}_2 \sin\theta_2 \cos\phi_2) \\
\dot{z}_2 &= -R_1\dot{\theta}_1 \sin\theta_1 - R_2\dot{\theta}_2 \sin\theta_2
\end{aligned}$$

[1] A convenient choice of coordinates is spherical polars, $r_1, \theta_1, \phi_1, r_2, \theta_2, \phi_2$, with the origin at the point of support and the polar axis pointing straight down. The four variables $\theta_1, \phi_1, \theta_2, \phi_2$ make a good choice of generalized coordinates.

1.3 Rewrite $\mathcal{L}$ in terms of $q_1, \cdots, q_n$ and $\dot{q}_1, \cdots, \dot{q}_n$.

$$\begin{aligned}\mathcal{L} = \frac{1}{2}(m_1+m_2)R_1^2(\dot{\theta}_1^2 + \dot{\phi}_1^2\sin^2\theta_1) + \frac{1}{2}m_2\Big\{2R_1R_2\Big[\dot{\theta}_1\dot{\theta}_2\big(\cos\theta_1\cos\theta_2\cos(\phi_1-\phi_2) + \sin\theta_1\sin\theta_2\big) \\ + \dot{\theta}_1\dot{\phi}_2\cos\theta_1\sin\theta_2\sin(\phi_1-\phi_2) + \dot{\phi}_1\dot{\theta}_2\sin\theta_1\cos\theta_2\sin(\phi_2-\phi_1) + \dot{\phi}_1\dot{\phi}_2\sin\theta_1\sin\theta_2\cos(\phi_1-\phi_2)\Big] \\ + R_2^2(\dot{\theta}_2^2 + \dot{\phi}_2^2\sin^2\theta_2)\Big\} + (m_1+m_2)gR_1\cos\theta_1 + m_2gR_2\cos\theta_2\end{aligned}$$

1.4 Write down the n Lagrange equations.

1.4.1 Equation of motion for θ_1

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}_1}\right) - \frac{\partial\mathcal{L}}{\partial\theta_1} = 0$$

$$\begin{aligned}(m_1+m_2)(R_1\ddot{\theta}_1 - R_1\dot{\phi}_1^2\sin\theta_1\cos\theta_1 + g\sin\theta_1) + m_2R_2\big\{\ddot{\theta}_2\big[\cos\theta_1\cos\theta_2\cos(\phi_1-\phi_2) + \sin\theta_1\sin\theta_2\big] \\ +\dot{\theta}_2\big[-\dot{\theta}_2\cos\theta_1\sin\theta_2\cos(\phi_1-\phi_2) - \cos\theta_1\cos\theta_2\sin(\phi_1-\phi_2)(\dot{\phi}_1-\dot{\phi}_2) + \dot{\theta}_2\sin\theta_1\cos\theta_2\big] \\ +\ddot{\phi}_2\cos\theta_1\sin\theta_2\sin(\phi_1-\phi_2) + \dot{\theta}_2\dot{\phi}_2\cos\theta_1\cos\theta_2\sin(\phi_1-\phi_2) + \dot{\phi}_2\cos\theta_1\sin\theta_2\cos(\phi_1-\phi_2)(\dot{\phi}_1-\dot{\phi}_2) \\ -\dot{\phi}_1\dot{\theta}_2\cos\theta_1\cos\theta_2\sin(\phi_2-\phi_1) - \dot{\phi}_1\dot{\phi}_2\cos\theta_1\sin\theta_2\cos(\phi_1-\phi_2)\big\} = 0\end{aligned}$$

1.4.2 Equation of motion for θ_2

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}_2}\right) - \frac{\partial\mathcal{L}}{\partial\theta_2} = 0$$

$$\begin{aligned}R_1\Big[\ddot{\theta}_1\big(\cos\theta_1\cos\theta_2\cos(\phi_1-\phi_2) + \sin\theta_1\sin\theta_2\big) + \dot{\theta}_1\big(-\dot{\theta}_1\sin\theta_1\cos\theta_2\cos(\phi_1-\phi_2) \\ -\cos\theta_1\cos\theta_2\sin(\phi_1-\phi_2)(\dot{\phi}_1-\dot{\phi}_2) + \dot{\theta}_1\cos\theta_1\sin\theta_2\big) + \ddot{\phi}_1\sin\theta_1\cos\theta_2\sin(\phi_2-\phi_1) \\ +\dot{\phi}_1\dot{\theta}_1\cos\theta_1\cos\theta_2\sin(\phi_2-\phi_1) - \dot{\phi}_1^2\sin\theta_1\cos\theta_2\cos(\phi_2-\phi_1) - \dot{\theta}_1\dot{\phi}_2\cos\theta_1\cos\theta_2\sin(\phi_1-\phi_2)\Big] \\ +R_2\big(\ddot{\theta}_2 - \dot{\phi}_2^2\sin\theta_2\cos\theta_2\big) + g\sin\theta_2 = 0\end{aligned}$$

1.4.3 Equation of motion for ϕ_1

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\phi}_1}\right) - \frac{\partial\mathcal{L}}{\partial\phi_1} = 0$$

$$\begin{aligned}(m_1+m_2)R_1(\ddot{\phi}_1\sin^2\theta_1 + 2\dot{\phi}_1\dot{\theta}_1\sin\theta_1\cos\theta_1) + m_2R_2\Big[\ddot{\theta}_2\sin\theta_1\cos\theta_2\sin(\phi_2-\phi_1) - \dot{\theta}_2^2\sin\theta_1\sin\theta_2\sin(\phi_2-\phi_1) \\ +2\dot{\theta}_2\dot{\phi}_2\sin\theta_1\cos\theta_2\cos(\phi_2-\phi_1) + \ddot{\phi}_2\sin\theta_1\sin\theta_2\cos(\phi_1-\phi_2) + \dot{\phi}_2^2\sin\theta_1\sin\theta_2\sin(\phi_1-\phi_2)\Big] = 0\end{aligned}$$

1.4.4 Equation of motion for ϕ_2

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\phi}_2}\right) - \frac{\partial\mathcal{L}}{\partial\phi_2} = 0$$

$$\begin{aligned}R_1\big[\ddot{\theta}_1\cos\theta_1\sin\theta_2\sin(\phi_1-\phi_2) - \dot{\theta}_1^2\sin\theta_1\sin\theta_2\sin(\phi_1-\phi_2) + 2\dot{\theta}_1\dot{\phi}_1\cos\theta_1\sin\theta_2\cos(\phi_1-\phi_2) \\ +\ddot{\phi}_1\sin\theta_1\sin\theta_2\cos(\phi_1-\phi_2) - \dot{\phi}_1^2\sin\theta_1\sin\theta_2\sin(\phi_1-\phi_2)\big] + R_2\big(\ddot{\phi}_2\sin^2\theta_2 + 2\dot{\theta}_2\dot{\phi}_2\sin\theta_2\cos\theta_2\big) = 0\end{aligned}$$

2 Explain what the ϕ equations tell us about the z component of angular momentum ℓ_z.

$$
\begin{aligned}
\dot{p}_{\phi_1} + \dot{p}_{\phi_2} &= \frac{\partial \mathcal{L}}{\partial \phi_1} + \frac{\partial \mathcal{L}}{\partial \phi_2} \\
\dot{p}_{\phi_1} + \dot{p}_{\phi_2} &= m_2 R_1 R_2 \big[-\dot{\theta}_1 \dot{\theta}_2 \cos\theta_1 \cos\theta_2 \sin(\phi_1 - \phi_2) + \dot{\theta}_1 \dot{\phi}_2 \cos\theta_1 \sin\theta_2 \cos(\phi_1 - \phi_2) \\
&\qquad - \dot{\phi}_1 \dot{\theta}_2 \sin\theta_1 \cos\theta_2 \cos(\phi_2 - \phi_1) - \dot{\phi}_1 \dot{\phi}_2 \sin\theta_1 \sin\theta_2 \sin(\phi_1 - \phi_2) \big] \\
&\qquad + m_2 R_1 R_2 \big[\dot{\theta}_1 \dot{\theta}_2 \cos\theta_1 \cos\theta_2 \sin(\phi_1 - \phi_2) - \dot{\theta}_1 \dot{\phi}_2 \cos\theta_1 \sin\theta_2 \cos(\phi_1 - \phi_2) \\
&\qquad + \dot{\theta}_2 \dot{\phi}_1 \sin\theta_1 \cos\theta_2 \cos(\phi_2 - \phi_1) + \dot{\phi}_1 \dot{\phi}_2 \sin\theta_1 \sin\theta_2 \sin(\phi_1 - \phi_2) \big] \\
\frac{d}{dt}[p_{\phi_1} + p_{\phi_2}] &= 0 \\
p_{\phi_1} + p_{\phi_2} &= \text{constant}
\end{aligned}
$$

The ϕ equations tell us that the z component of angular momentum, $\ell_z = p_{\phi_1} + p_{\phi_2}$ is constant.

3 For the special case that ϕ_i = const, describe what the θ equations tell us.

If ϕ_1, ϕ_2 is constant, the θ equations reduce to

$$(m_1 + m_2)R_1(R_1\ddot{\theta}_1 + g\sin\theta_1) + m_2R_1R_2\big\{\ddot{\theta}_2\cos(\theta_1 - \theta_2) + \dot{\theta}_2^2\sin(\theta_1 - \theta_2)\big\} = 0,$$
$$m_2R_2\big\{R_1\big[\ddot{\theta}_1\cos(\theta_1 - \theta_2) - \dot{\theta}_1^2\sin(\theta_1 - \theta_2)\big] + R_2\ddot{\theta}_2 + g\sin\theta_2\big\} = 0,$$

which are the equations for a planar double pendulum. That is, in this case, the double pendulum swings in a single vertical plane, $\phi = \phi_o$, just like a simple double pendulum.

4 Discuss the existence of angles $\theta_{i,0}$ at which θ_i can remain constant. Why is this motion called a conical pendulum?

Now θ_i can only remain constant if and only if $\dot{\theta}_i = \ddot{\theta}_i = 0$, and (by definition of a conical pendulum) if $\dot{\phi}_1 = \dot{\phi}_2 \equiv \dot{\Phi}$ remains constant and equal then $\ddot{\phi}_i = 0$, which requires that

$$\begin{aligned}(m_1+m_2)(g\sin\theta_{1,0} - R_1\dot{\Phi}^2\sin\theta_{1,0}\cos\theta_{1,0}) - m_2R_2\dot{\Phi}^2\cos\theta_{1,0}\sin\theta_{2,0}\cos(\phi_1-\phi_2) &= 0,\\ -R_1\dot{\Phi}^2\sin\theta_{1,0}\cos\theta_{2,0}\cos(\phi_2-\phi_1) - R_2\dot{\Phi}^2\sin\theta_{2,0}\cos\theta_{2,0} + g\sin\theta_{2,0} &= 0,\\ m_2R_2\dot{\Phi}^2\sin\theta_{1,0}\sin\theta_{2,0}\sin(\phi_1-\phi_2) &= 0,\\ -R_1\dot{\Phi}^2\sin\theta_{1,0}\sin\theta_{2,0}\sin(\phi_1-\phi_2) &= 0.\end{aligned}$$

The third and fourth equations of motion are redundant under this special case. For a nontrivial conical motion, $\dot{\Phi} \neq 0$, $\sin\theta_i \neq 0$, so it forces

$$\phi_1 - \phi_2 = n\pi.$$

In other words, the two links must have their horizontal projections either aligned $(\phi_1 - \phi_2) = 0$ or anti-aligned $(\phi_1 - \phi_2) = \pi$. Taking the aligned branch $(\phi_1 - \phi_2) = 0$ leaves

$$\begin{aligned}(m_1+m_2)(g\sin\theta_{1,0} - R_1\dot{\Phi}^2\sin\theta_{1,0}\cos\theta_{1,0}) - m_2R_2\dot{\Phi}^2\cos\theta_{1,0}\sin\theta_{2,0} &= 0,\\ -R_1\dot{\Phi}^2\sin\theta_{1,0}\cos\theta_{2,0} - R_2\dot{\Phi}^2\sin\theta_{2,0}\cos\theta_{2,0} + g\sin\theta_{2,0} &= 0.\end{aligned}$$

These are two equations for three unknowns $\theta_{1,0}$, $\theta_{2,0}$, $\dot{\Phi}$. This means we expect a family of double-conical solutions rather than one unique configuration. We can solve the second equation for $\dot{\Phi}^2$ analytically:

$$\dot{\Phi}^2 = \frac{g\tan\theta_{2,0}}{R_1\sin\theta_{1,0} + R_2\sin\theta_{2,0}}.$$

Substituting this expression for $\dot{\Phi}^2$ into the first equation gives

$$(m_1+m_2)\sin\theta_{1,0} = \frac{[(m_1+m_2)R_1\sin\theta_{1,0} + m_2R_2\sin\theta_{2,0}]\cos\theta_{1,0}\tan\theta_{2,0}}{R_1\sin\theta_{1,0} + R_2\sin\theta_{2,0}}.$$

This equation[2] tells us which pairs $(\theta_{1,0}, \theta_{2,0})$ can form an aligned double-conical pendulum. Once we choose a point on that curve, the equation above gives the corresponding $\dot{\Phi}$.
In this motion the upper rod of the pendulum traces out a vertical cone of half angle $\theta_{1,0}$. The lower rod maintains a constant inclination while its attachment at the upper bob moves. Relative to that translating attachment, it also sweeps out a cone.
Taking the anti-aligned branch$(\phi_1 - \phi_2) = \pi$ leaves

$$\begin{aligned}(m_1+m_2)(g\sin\theta_{1,0} - R_1\dot{\Phi}^2\sin\theta_{1,0}\cos\theta_{1,0}) + m_2R_2\dot{\Phi}^2\cos\theta_{1,0}\sin\theta_{2,0} &= 0,\\ R_1\dot{\Phi}^2\sin\theta_{1,0}\cos\theta_{2,0} - R_2\dot{\Phi}^2\sin\theta_{2,0}\cos\theta_{2,0} + g\sin\theta_{2,0} &= 0.\end{aligned}$$

Similarly, we can solve the second equation for $\dot{\Phi}^2$ analytically:

$$\dot{\Phi}^2 = \frac{g\tan\theta_{2,0}}{R_2\sin\theta_{2,0} - R_1\sin\theta_{1,0}}.$$

Substituting this expression for $\dot{\Phi}^2$ into the first equation gives

$$(m_1+m_2)\sin\theta_{1,0} = \frac{[(m_1+m_2)R_1\sin\theta_{1,0} - m_2R_2\sin\theta_{2,0}]\cos\theta_{1,0}\tan\theta_{2,0}}{R_2\sin\theta_{2,0} - R_1\sin\theta_{1,0}}.$$

[2]These expressions apply where the denominators are nonzero and $\cos\theta_{2,0} \neq 0$. A nontrivial rotating solution also requires $\dot{\Phi} > 0$. Cases excluded by these must be examined using the original equilibrium equations.

5 Show that if $\theta_i = \theta_{i,0} + \epsilon_i$ and $\phi_i = \dot{\phi}t + \phi_0 + \eta_i$, with ϵ, η small, then θ_i oscillates about $\theta_{i,0}$ in harmonic motion. Describe the motion of the pendulum's bobs.

Full nonlinear perturbated EOMs for the aligned equilibrium:

$$
\begin{aligned}
&(m_1+m_2)[R_1\ddot{\epsilon}_1 - R_1(\dot{\Phi}+\dot{\eta}_1)^2\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{1,0}+\epsilon_1) + g\sin(\theta_{1,0}+\epsilon_1)]\\
&+m_2R_2\big\{\ddot{\epsilon}_2\big[\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)+\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\big]\\
&+\dot{\epsilon}_2\big[-\dot{\epsilon}_2\cos(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)-\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)(\dot{\eta}_1-\dot{\eta}_2)\\
&+\dot{\epsilon}_2\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\big]+\ddot{\eta}_2\cos(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)\\
&+\dot{\epsilon}_2(\dot{\Phi}+\dot{\eta}_2)\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)+(\dot{\Phi}+\dot{\eta}_2)\cos(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)(\dot{\eta}_1-\dot{\eta}_2)\\
&-(\dot{\Phi}+\dot{\eta}_1)\dot{\epsilon}_2\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_2-\eta_1)-(\dot{\Phi}+\dot{\eta}_1)(\dot{\Phi}+\dot{\eta}_2)\cos(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)\big\}=0
\end{aligned}
$$

$$
\begin{aligned}
&R_1\big\{\ddot{\epsilon}_1\left[\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)+\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\right]\\
&+\dot{\epsilon}_1\big[-\dot{\epsilon}_1\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)-\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)(\dot{\eta}_1-\dot{\eta}_2)\\
&+\dot{\epsilon}_1\cos(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\big]+\ddot{\eta}_1\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_2-\eta_1)\\
&+(\dot{\Phi}+\dot{\eta}_1)\dot{\epsilon}_1\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_2-\eta_1)-(\dot{\Phi}+\dot{\eta}_1)^2\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\cos(\eta_2-\eta_1)\\
&-\dot{\epsilon}_1(\dot{\Phi}+\dot{\eta}_2)\cos(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)\big\}+R_2\left[\ddot{\epsilon}_2-(\dot{\Phi}+\dot{\eta}_2)^2\sin(\theta_{2,0}+\epsilon_2)\cos(\theta_{2,0}+\epsilon_2)\right]\\
&+g\sin(\theta_{2,0}+\epsilon_2)=0
\end{aligned}
$$

$$
\begin{aligned}
&(m_1+m_2)R_1[\ddot{\eta}_1\sin^2(\theta_{1,0}+\epsilon_1)+2(\dot{\Phi}+\dot{\eta}_1)\dot{\epsilon}_1\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{1,0}+\epsilon_1)]\\
&+m_2R_2\big[\ddot{\epsilon}_2\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\sin(\eta_2-\eta_1)-\dot{\epsilon}_2^2\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\sin(\eta_2-\eta_1)\\
&+2\dot{\epsilon}_2(\dot{\Phi}+\dot{\eta}_2)\sin(\theta_{1,0}+\epsilon_1)\cos(\theta_{2,0}+\epsilon_2)\cos(\eta_2-\eta_1)+\ddot{\eta}_2\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)\\
&+(\dot{\Phi}+\dot{\eta}_2)^2\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)\big]=0
\end{aligned}
$$

$$
\begin{aligned}
&R_1\big[\ddot{\epsilon}_1\cos(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)-\dot{\epsilon}_1^2\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)\\
&+2\dot{\epsilon}_1(\dot{\Phi}+\dot{\eta}_1)\cos(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)+\ddot{\eta}_1\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\cos(\eta_1-\eta_2)\\
&-(\dot{\Phi}+\dot{\eta}_1)^2\sin(\theta_{1,0}+\epsilon_1)\sin(\theta_{2,0}+\epsilon_2)\sin(\eta_1-\eta_2)\big]\\
&+R_2\big[\ddot{\eta}_2\sin^2(\theta_{2,0}+\epsilon_2)+2\dot{\epsilon}_2(\dot{\Phi}+\dot{\eta}_2)\sin(\theta_{2,0}+\epsilon_2)\cos(\theta_{2,0}+\epsilon_2)\big]=0
\end{aligned}
$$

Linearized approximation:
If θ_i is close to the equilibrium value, $\theta_i = \theta_{i,0} + \epsilon_i$ and if ϕ_i is close to the equilibrium value, $\phi_i = \dot{\Phi}t + \phi_{i,0} + \eta_i$, we can write

$$
\begin{aligned}
\sin(\theta_{i,0}+\epsilon_i) &= \sum_{n=0}^{\infty}\frac{d^n}{d\epsilon_i^n}[\sin(\theta_{i,0}+\epsilon_i)]_{\epsilon_i=0}\frac{\epsilon_i^n}{n!} \approx \sin\theta_{i,0}+\epsilon_i\cos\theta_{i,0}\\
\cos(\theta_{i,0}+\epsilon_i) &= \sum_{n=0}^{\infty}\frac{d^n}{d\epsilon_i^n}[\cos(\theta_{i,0}+\epsilon_i)]_{\epsilon_i=0}\frac{\epsilon_i^n}{n!} \approx \cos\theta_{i,0}-\epsilon_i\sin\theta_{i,0}\\
\sin(\eta_1-\eta_2) &= \sum_{n=0}^{\infty}\left(\frac{\partial^n}{\partial\eta_1^n}[\sin(\eta_1-\eta_2)]_{\eta_i=0}\frac{\eta_1^n}{n!}+\frac{\partial^n}{\partial\eta_2}[\sin(\eta_1-\eta_2)]_{\eta_i=0}\frac{\eta_2^n}{n!}\right)\approx \eta_1-\eta_2\\
\cos(\eta_1-\eta_2) &= \sum_{n=0}^{\infty}\left(\frac{\partial^n}{\partial\eta_1}[\cos(\eta_1-\eta_2)]_{\eta_i=0}\frac{\eta_1^n}{n!}+\frac{\partial^n}{\partial\eta_2}[\cos(\eta_1-\eta_2)]_{\eta_i=0}\frac{\eta_2^n}{n!}\right)\approx 1
\end{aligned}
$$

$$(m_1+m_2)[R_1\ddot{\epsilon}_1 - R_1\dot{\Phi}^2(\sin\theta_{1,0}\cos\theta_{1,0} - \epsilon_1\sin^2\theta_{1,0} + \epsilon_1\cos^2\theta_{1,0}) - 2R_1\dot{\Phi}\dot{\eta}_1\sin\theta_{1,0}\cos\theta_{1,0} + g(\sin\theta_{1,0} + \epsilon_1\cos\theta_{1,0})]$$
$$+m_2R_2\big[\ddot{\epsilon}_2(\cos\theta_{1,0}\cos\theta_{2,0} + \sin\theta_{1,0}\sin\theta_{2,0}) - \dot{\Phi}^2(\cos\theta_{1,0}\sin\theta_{2,0} + \epsilon_2\cos\theta_{1,0}\cos\theta_{2,0} - \epsilon_1\sin\theta_{1,0}\sin\theta_{2,0})$$
$$-2\dot{\Phi}\dot{\eta}_2\cos\theta_{1,0}\sin\theta_{2,0}\big] = 0$$

$$R_1\big[\ddot{\epsilon}_1(\cos\theta_{1,0}\cos\theta_{2,0} + \sin\theta_{1,0}\sin\theta_{2,0}) - \dot{\Phi}^2(\sin\theta_{1,0}\cos\theta_{2,0} - \epsilon_2\sin\theta_{1,0}\sin\theta_{2,0} + \epsilon_1\cos\theta_{1,0}\cos\theta_{2,0})$$
$$-2\dot{\Phi}\dot{\eta}_1\sin\theta_{1,0}\cos\theta_{2,0}\big] + R_2\big[\ddot{\epsilon}_2 - \dot{\Phi}^2\big(\sin\theta_{2,0}\cos\theta_{2,0} - \epsilon_2\sin^2\theta_{2,0} + \epsilon_2\cos^2\theta_{2,0}\big) - 2\dot{\Phi}\dot{\eta}_2\sin\theta_{2,0}\cos\theta_{2,0}\big]$$
$$+g(\sin\theta_{2,0} + \epsilon_2\cos\theta_{2,0}) = 0$$

$$(m_1+m_2)R_1(\ddot{\eta}_1\sin^2\theta_{1,0} + 2\dot{\Phi}\dot{\epsilon}_1\sin\theta_{1,0}\cos\theta_{1,0})$$
$$+m_2R_2\big[2\dot{\Phi}\dot{\epsilon}_2\sin\theta_{1,0}\cos\theta_{2,0} + \ddot{\eta}_2\sin\theta_{1,0}\sin\theta_{2,0} + \dot{\Phi}^2\sin\theta_{1,0}\sin\theta_{2,0}(\eta_1-\eta_2)\big] = 0$$

$$R_1\big[2\dot{\Phi}\dot{\epsilon}_1\cos\theta_{1,0}\sin\theta_{2,0} + \ddot{\eta}_1\sin\theta_{1,0}\sin\theta_{2,0} - \dot{\Phi}^2\sin\theta_{1,0}\sin\theta_{2,0}(\eta_1-\eta_2)\big] + R_2\big[\ddot{\eta}_2\sin^2\theta_{2,0} + 2\dot{\Phi}\dot{\epsilon}_2\sin\theta_{2,0}\cos\theta_{2,0}\big] = 0$$

The aligned branch condition from section 4 simplifies the first two equations.

$$(m_1+m_2)[R_1\ddot{\epsilon}_1 - R_1\dot{\Phi}^2\epsilon_1(\cos^2\theta_{1,0} - \sin^2\theta_{1,0}) - 2R_1\dot{\Phi}\dot{\eta}_1\sin\theta_{1,0}\cos\theta_{1,0} + g\epsilon_1\cos\theta_{1,0}]$$
$$+m_2R_2\big[\ddot{\epsilon}_2(\cos\theta_{1,0}\cos\theta_{2,0} + \sin\theta_{1,0}\sin\theta_{2,0}) - \dot{\Phi}^2(\epsilon_2\cos\theta_{1,0}\cos\theta_{2,0} - \epsilon_1\sin\theta_{1,0}\sin\theta_{2,0}) - 2\dot{\Phi}\dot{\eta}_2\cos\theta_{1,0}\sin\theta_{2,0}\big] = 0$$

$$R_1\big[\ddot{\epsilon}_1(\cos\theta_{1,0}\cos\theta_{2,0} + \sin\theta_{1,0}\sin\theta_{2,0}) - \dot{\Phi}^2(\epsilon_1\cos\theta_{1,0}\cos\theta_{2,0} - \epsilon_2\sin\theta_{1,0}\sin\theta_{2,0}) - 2\dot{\Phi}\dot{\eta}_1\sin\theta_{1,0}\cos\theta_{2,0}\big]$$
$$+R_2[\ddot{\epsilon}_2 - \dot{\Phi}^2\epsilon_2(\cos^2\theta_{2,0} - \sin^2\theta_{2,0}) - 2\dot{\Phi}\dot{\eta}_2\sin\theta_{2,0}\cos\theta_{2,0}] + g\epsilon_2\cos\theta_{2,0} = 0$$

Written in the compact matrix form,

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{B}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = 0, \qquad \mathbf{q} = (\epsilon_1, \epsilon_2, \eta_1, \eta_2)^T$$

Trying to find solutions of this form,

$$\mathbf{q}(t) = \mathbf{a}e^{i\omega t}$$

we obtain

$$(\mathbf{K} - \omega^2\mathbf{M} + i\omega\mathbf{B})\mathbf{a} = 0.$$

If

$$\det(\mathbf{K} - \omega^2\mathbf{M} + i\omega\mathbf{B}) = 0,$$

then there is a nontrivial solution.